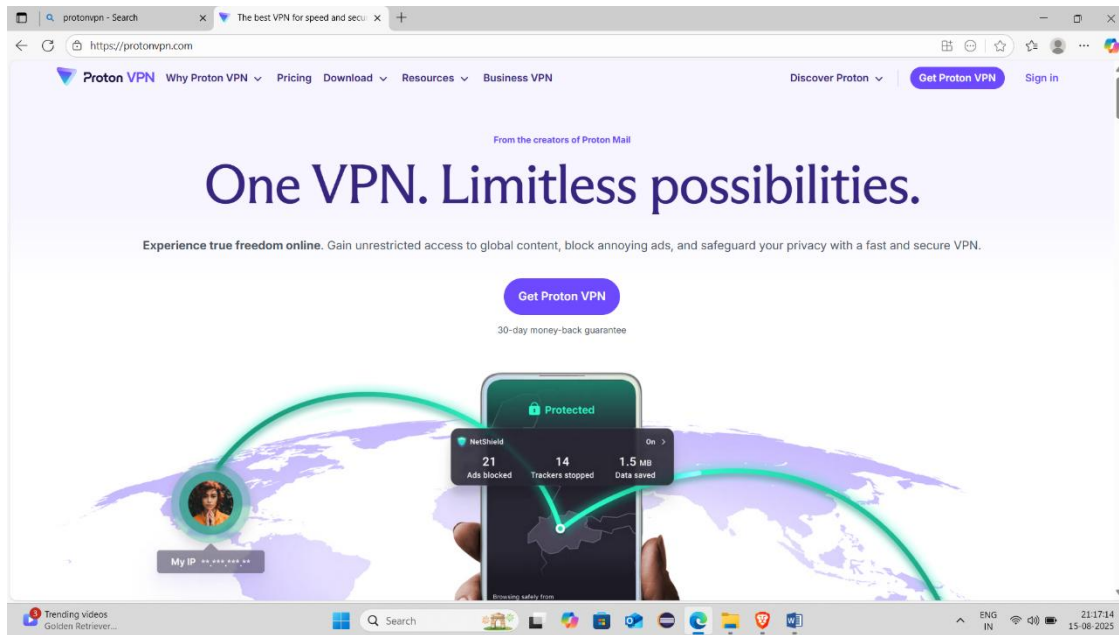


Task 8: Working With VPNs

Step 1: Choose a Reputable Free VPN

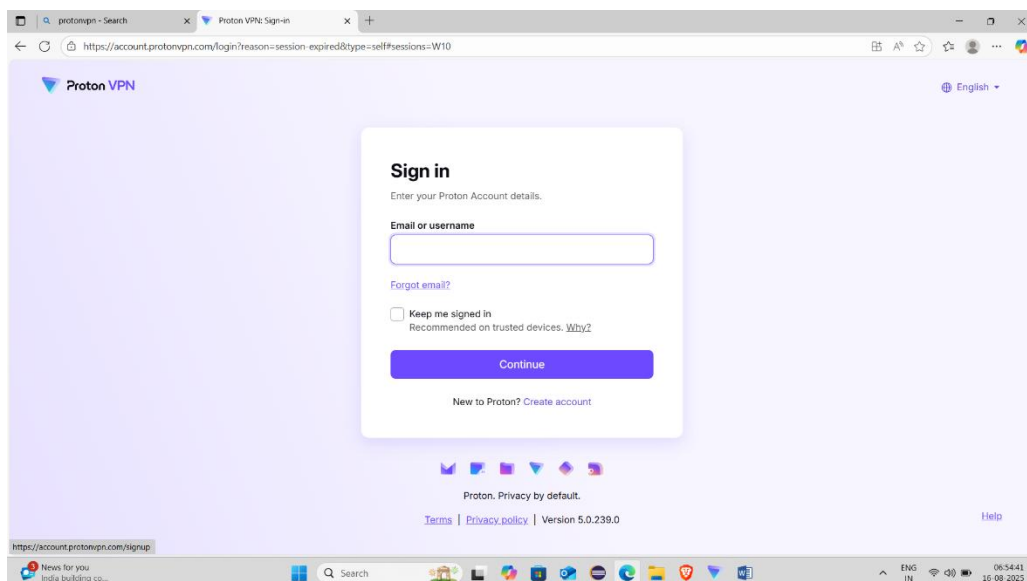
Pick a well-known free VPN (safe for learning purposes):

- **ProtonVPN Free** – No data limit, high trust
- Avoid random unknown VPNs – can be dangerous.



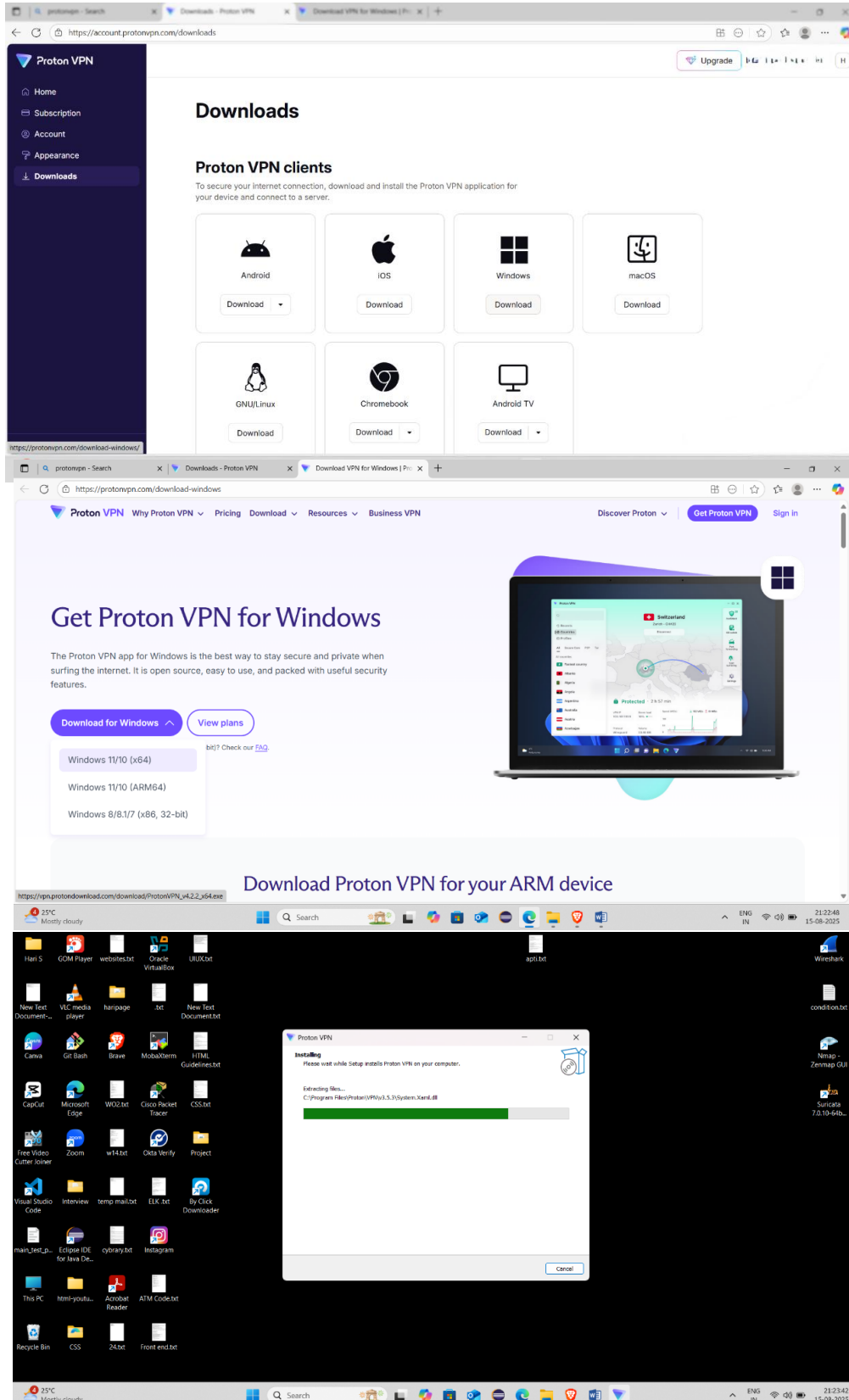
Step 2: Sign Up

- Go to the VPN's official website (example: protonvpn.com).
- Create a free account with your email.
- Confirm verification email (if required).



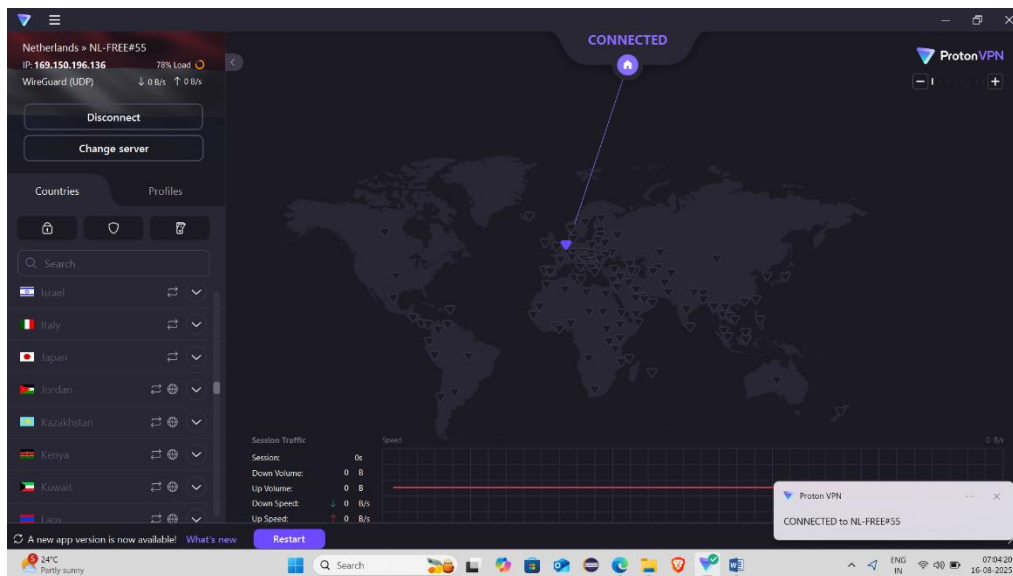
Step 3: Download and Install the VPN Client

- Download the VPN app for your OS (Windows / Android / iOS).
- Follow installation steps.
- Log in using the account you created.



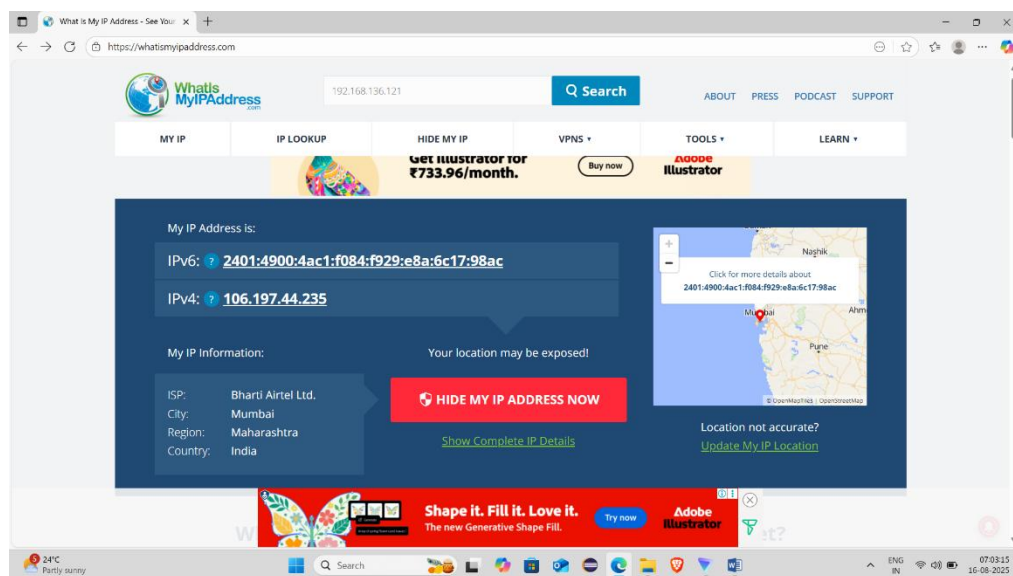
Step 4: Connect to a Server

- Open the VPN app.
- Select a **server location** (choose the closest one for better speed).
- Click **Connect**.
- Wait until it shows “Connected”.

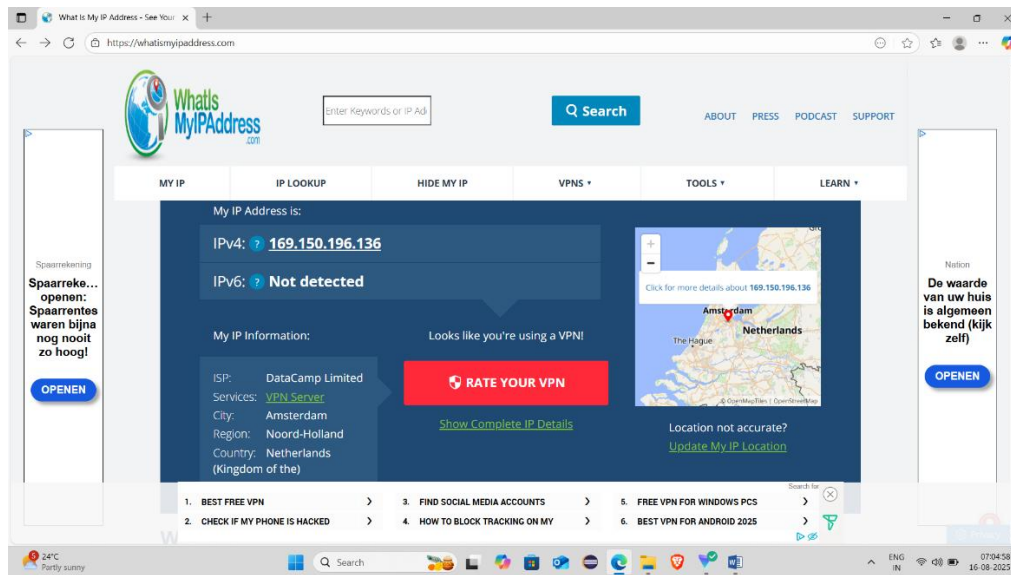


Step 5: Verify Your IP Has Changed

- Before connecting, note your original IP at: <https://www.whatismyipaddress.com>

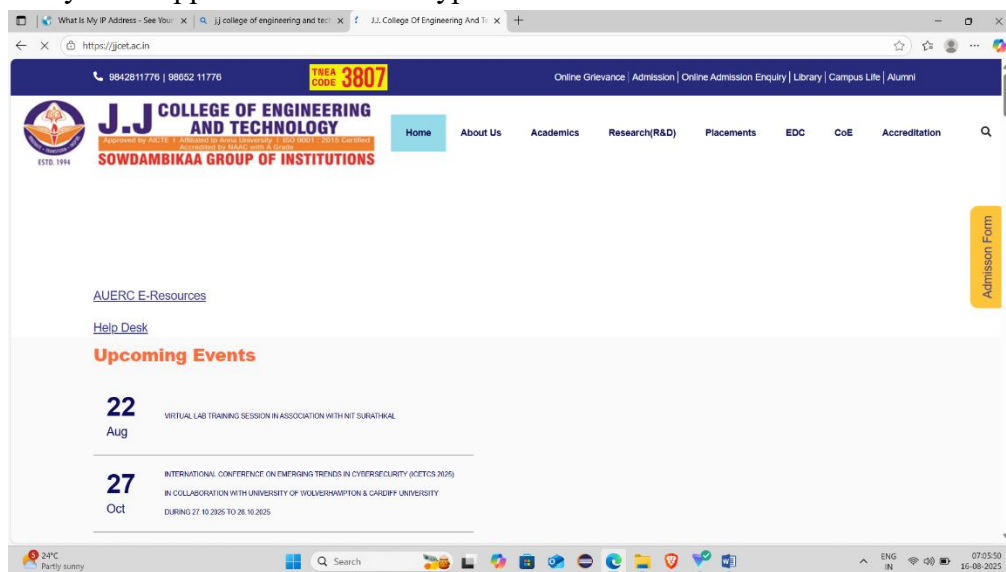


- After connecting, refresh the website → check if IP & location changed.
- Write both down as evidence.



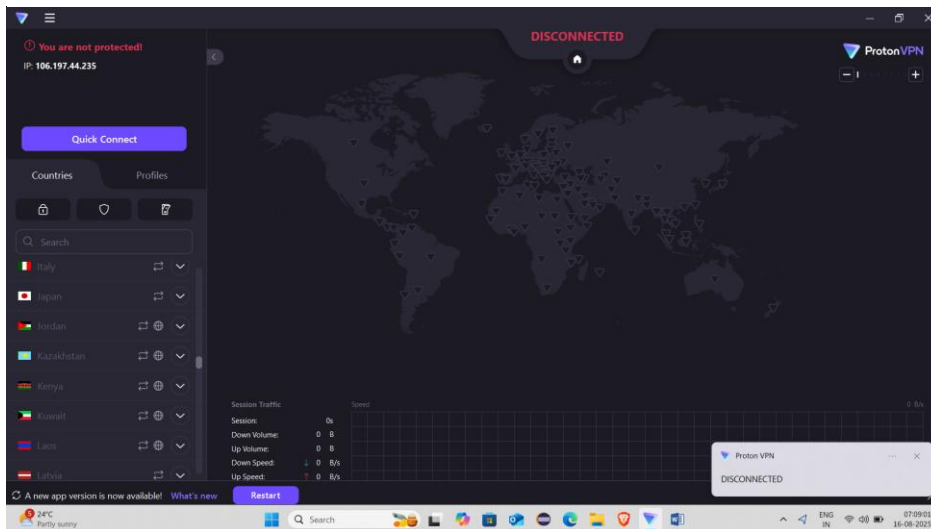
Step 6: Test Browsing & Encryption

- Open any website (e.g., <https://example.com> or your college site).
- Check if it loads normally → that means traffic is going through VPN.
- Many VPN apps also show “Encrypted” or “Secure Tunnel Active”.



Step 7: Disconnect VPN & Compare

- Disconnect from the VPN using the app.
- Wait till status is "Disconnected".
- Check your IP again → it should show your real IP now.
- Load the same website again and see if speed differs.



Step 8: Research VPN Encryption & Privacy

- What is AES-256 encryption

AES 256 Encryption 1 2

AES (Advanced Encryption Standard) is a widely used symmetric encryption algorithm designed to secure data by converting it into an unreadable format without the proper decryption key. AES-256, in particular, uses a **256-bit key**, offering a high level of security suitable for protecting sensitive information.

Key Features of AES-256

- **Block Cipher:** Operates on fixed-size blocks of 128 bits.
- **Key Length:** Uses a 256-bit key, providing stronger encryption compared to 128-bit or 192-bit keys.
- **Rounds:** Performs 14 rounds of encryption, where each round includes substitution, permutation, and mixing operations.
- **Symmetric Encryption:** The same key is used for both encryption and decryption, making it faster than asymmetric encryption.

How AES-256 Works

AES-256 encrypts data in blocks of 128 bits using the following steps:

1. **SubBytes:** Each byte is substituted using a predefined substitution box (S-box).
2. **ShiftRows:** Rows of the 4x4 byte matrix are shifted by varying offsets.
3. **MixColumns:** Columns are mixed using matrix multiplication (skipped in the final round).
4. **AddRoundKey:** The result is XORed with a round-specific key derived from the main key.

The process is repeated for 14 rounds, ensuring robust encryption. Decryption reverses these steps using inverse operations.

- Difference Between Proxy & VPN.

Proxy vs VPN: Key Differences, Benefits, and When to Use Each

- **Proxy:** Acts as a gateway between the internet and users, providing traffic source anonymization and potentially supporting traffic distribution. It is less expensive than VPN services and is suitable for basic browsing tasks. [↔ 5](#)
- **VPN:** Provides an encrypted tunnel through which the web surfing data runs, offering a higher level of security and privacy. It is more expensive than proxies but is ideal for handling sensitive data, secure browsing, online privacy, and protection against hackers. [↔ 5](#)

In summary, **VPNs are better for privacy and security**, while **proxies are useful for light browsing tasks that don't involve sensitive data**. [↔ 1](#)

[↔ 5 Sources](#)

- Pros & Cons Of VPN.

VPN

PROS AND CONS

Everything you need to know before purchasing a VPN

PROS	CONS
 Unblock Netflix and Access Censored Content Unblock all your favorite streaming services, such as Netflix, Hulu, and BBC	 Slow Connection Speed Be sure to get a VPN with fast speeds, so you can stream with no buffering or lagging
 Keeps You Safe A VPN secures your connection and protect you from hackers and other online threats	 Anti-VPN Software Choose a VPN that can get past the VPN blocks used by Netflix and other streaming services
 Saves You Money Mask your IP address and save some serious cash when booking flights or hotel rooms	 Connection Can Drop Your VPN should have a kill switch, to prevent accidental IP leaks
 Affordable Security VPNs are significantly cheaper and easier to set up than other security plans	 Can Be Difficult to Configure Several providers offer easy-to-use, beginner-friendly interfaces



- protonvpn.com/blog/vpn-encryption

ProtonVPN encryption features

ProtonVPN employs robust encryption protocols to protect user data, ensuring a secure and private browsing experience. Here are some of the key encryption features:

- **AES-256 and ChaCha20:** ProtonVPN uses these ciphers for encryption, with AES-256 being the most secure option. ChaCha20 is also used for WireGuard VPN protocol connections.  1
- **Perfect Forward Secrecy:** This feature ensures that each session generates a unique set of encryption keys, keeping past sessions secure even if future sessions are compromised.  1
- **Diffie-Hellman Key Exchange:** This method securely generates session keys without communicating them, enhancing the security of the VPN connection.  1
- **Open Source:** ProtonVPN's apps are open source and available for public inspection, allowing the cybersecurity community to find and fix potential vulnerabilities.  1
- **Independent Audits:** All ProtonVPN apps undergo independent audits by leading cybersecurity firms to ensure they perform as advertised.  1
ProtonVPN's commitment to strong encryption and privacy makes it a reliable choice for users seeking to protect their online activities.  4

Useful Resources / Links:

Purpose	Link / Resource
IP Check	https://whatismyipaddress.com
VPN Provider	https://protonvpn.com or https://windscribe.com
VPN Encryption Info	https://protonvpn.com/blog/vpn-encryption/